

On the Access Time in Mobile Hybrid Networks

Haiyang Zheng and Wenye Wang
 Department of Electrical and Computer Engineering
 North Carolina State University
 Raleigh, NC27606
 Email: {hzheng5, wwang}@ncsu.edu

Abstract—This paper investigates the *access time* of mobile nodes to infrastructure networks in Mobile Hybrid Networks (MHNs), e.g., sensor-actuator networks, where mobile nodes move around the coexisting infrastructure networks. In such networks, mobile nodes may relay data packets in a hop-by-hop fashion, and eventually deliver packets to the wired networks. Our objective is to study the lower bound of the access time, which implies the minimum time to deliver data packets from low-speed, unreliable wireless networks to the high-speed, reliable infrastructure network. In particular, we propose a theoretical framework to analyze the packet propagation speed in MHNs, and to show that there is a unified lower bound on the access time regardless of data forwarding schemes. When wireless nodes move at an average speed of v and variance σ^2 towards an AP, and their density λ is small (e.g., the network is sparse and surely disconnected), we prove that the expected access time is lower bounded by $\frac{(L_b(0)-r)^2}{\sigma^2}$ if packet mobility is unbiased and $\frac{L_b(0)-r}{v}$ for the biased case, where $L_b(0)$ is the distance from the source node to the nearest access point. Here, packet mobility is the mobility of a packet due to both node mobility and data transmissions. We further propose a routing scheme to achieve such theoretical bounds.

I. INTRODUCTION

In recent years, there is a tremendous growth in the applications of multi-hop wireless networks, such as wildlife tracking and habitat monitoring sensor networks [1], military networks [2], and vehicular ad hoc networks (VANETs) [3]. Interestingly enough, it is clear that all these networks follow a similar paradigm. That is, wireless nodes may relay data packets in a hop-by-hop fashion, and eventually deliver packets to the wired networks. An interesting, yet important question is how to find a path from wireless nodes to the infrastructure networks in order to minimize the access time. Obviously, the shorter the path, in both time and space domain, the better services such networks can provide.

Many efforts have been made to understand fundamental properties and performance limits of wireless multi-hop networks under the implicit assumption that connectivity has been maintained (e.g., [4]–[7]). However, in many real systems, such as sensor-actuator networks, military networks, and vehicular ad hoc networks (VANETs), the network connectivity is not always guaranteed. Instead, intermittent connectivity may arise due to sparseness [8], [9], powering off of wireless nodes (e.g., save battery) [1], high mobility [3], or even for covertness [2]. While the performance of networks which are always connected has been well studied, little is known about the performance of

networks whose connectivity is intermittent (e.g., [10]–[16]), such as MHNs.

In this paper, we study the *access time* from wireless nodes to wired networks in MHNs. It is worthy of noting that this problem can be very challenging, because oftentimes, there is no end-to-end path in such networks, and conventional ad hoc routing schemes cannot be applied. However this does not mean that packets can never be delivered under intermittent connectivity. A packet can be sent over an existing link, buffered at the next hop until the next link toward the destination is available. In this way, data packets are either transmitted or carried by mobile nodes (requiring a store-carry-and forward routing model). Thus, a “path” is an alternation of packet transmissions and carriages. Our goal is to find the shortest path that connects any mobile user to the infrastructure network.

Particularly, we consider a network with mobile nodes moving in a square region, coexisting with an infrastructure node placed at the center of the network to help mobile nodes forward data packets. In order to study the properties of MHNs that are relevant to the field of applications, we are interested in sparse networks in which the node density λ is small. Indeed, most applications for MHNs are required to work as sparse mobile ad hoc networks (e.g., [11], [17]), where intermittent connectivity is due to node mobility and limited radio coverage.

As such networks may not be connected over time and there are a variety of routing schemes, it is impossible to follow the conventional approach, which takes node mobility and routing schemes in deriving theoretical limits of network performance. Instead, we consider *packet mobility*, i.e., the mobility or the paths of each packet in our analytical framework. Specifically, we define and follow the random process of packet mobility, and thus find the total time that takes one packet from a mobile node to the access point. Therefore, by using this new probabilistic model of packet delivery in mobile hybrid networks, we obtain the lower bound of the access time for any forwarding algorithm in mobile hybrid networks, which depends on the network size and packet mobility models. Particularly, we prove that the expected access time is lower bounded by $\frac{(L_b(0)-r)^2}{\sigma^2}$ if packet mobility is unbiased and $\frac{L_b(0)-r}{v}$ for the biased mobility, where $L_b(0)$ is the distance from the source node to the nearest access point.

The rest of the paper is organized as follows. In Section II, we describe the network model and formulate the *access time* problem. The theoretical bounds of access time under different

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packet mobility model are derived in Section III. A biased routing scheme has been proposed to achieve such theoretical bounds in Section IV. Simulations validate these analytical results in Section V, followed by conclusions in Section VI.

II. NETWORK MODEL

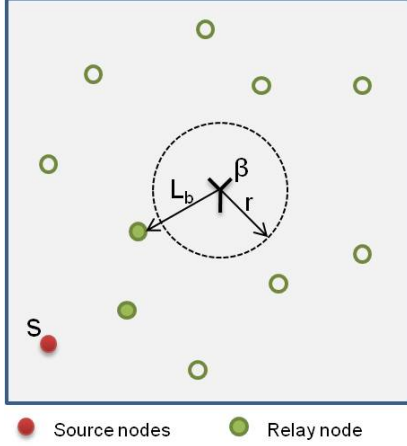


Fig. 1. Network Model.

In our network model shown in Figure 1, a traditional mobile ad hoc network is overlaid with an infrastructure networks. We assume that there is a set $\mathcal{A} = \{\alpha_1, \alpha_2, \dots\}$ of ad-hoc nodes, called α -nodes, randomly sparsely distributed in a square region Ω with length l . In addition, an infrastructure nodes, called β -node, is place at the center of the square region.

Initially, the α -nodes are distribute uniformly at random. Every α -node moves according to a stochastic mobile mode. Let $X_i(t)$ be the random variable that denotes the position of α -node i at time slot t . Assume at the beginning of every time slot, the nodes will estimate the distance to the nearest β -node and the step length towards, e.g. the distance it has moved (this can be done by measuring the signal strength from a centering β -node).

Since we are interested in computing analytical bounds on the access time, we do not consider here the effects of buffering or congestion. In other words, we assume that there is no contention for buffer and bandwidth. Although contention is an important factor for dense networks, we argue that it is significantly less of an issue for sparse networks.

To proceed, we'd like to present the following definition:

Definition 1: (Access Time): Fix any scheduling policy and fix a packet b that enters into the system at time slot 0. We define the following random variables for each packet b that needs to be communicated from its source node s to the β -nodes. Denote $\mathcal{R}_b(t)$ as the set of nodes that have received the packet by time $t \geq 0$. We call these nodes the “custodians” of the packet. Note that under our network model, the relay nodes are randomly selected because each node has a set of candidate next-hop nodes and does not know which of them it will meet to forward the packet. Among these mobile relays, there is one

mobile relay whose distance to the destination of packet b is the smallest. Let $L_b(t)$ denote this minimum distance, then

$$L_b(t) = \min\{\|X_i(t) - X_\beta\| : i \in \mathcal{R}_b(t) \text{ and } \beta \in \mathcal{B}\}.$$

The access time is then defined as

$$T_{up} = \inf_{t \geq 0} \{t : L_b(t) \leq r\}.$$

III. THE ACCESS TIME UNDER DIFFERENT PACKET MOBILITY MODELS

In this section, we will analyze the access time in MHNs, under different packet mobility models. Different from previous works, we here consider packet mobility, i.e., the combinations of both node mobility and data transmissions. In this way, we can ignore the details of a specific node mobility.

The following theorems present our generic lower bound on the access time in terms of different packet mobility models.

1) *Unbiased Packet Mobility*: The packet mobility is called *unbiased* if the corresponding sequence of random variable $L_b(t)$, $t \geq 1$, form a martingale defined on a filtered probability space $(\Omega, \mathcal{F}, P)$, e.g.,

$$E[L_b(t)|\mathcal{F}_{t-1}] = L_b(t-1).$$

Let $v_t = L_b(t) - L_b(t-1)$, $t \geq 1$ be the steps of the packet. We have

$$E[v_t|\mathcal{F}_{t-1}] = E[L_b(t-1) - L_b(t)|\mathcal{F}_{t-1}] = 0.$$

Define the random variables σ_t^2 by

$$\sigma_t^2 = E[v_t^2|\mathcal{F}_{t-1}] - E^2[v_t|\mathcal{F}_{t-1}].$$

With the above definitions, we are now ready to present the following theorem.

Theorem 1: Given any packet mobility model that is unbiased, the expected access time is lower bounded by $\frac{(L_b(0)-r)^2}{\sigma^2}$.

Proof: Let $Y_t = (L_b(0) - L_b(t))^2 - S_t$, where $L_b(t)$ is a martingale and $S_t = \sum_{i=1}^t \sigma_i^2$. We have,

$$\begin{aligned} E[Y_t|\mathcal{F}_{t-1}] &= E[(L_b(0) - L_b(t))^2 - S_t|\mathcal{F}_{t-1}] \\ &= E[(L_b(0) - L_b(t-1))^2 - S_{t-1} + v_t^2 \\ &\quad + 2v_t(L_b(0) - L_b(t-1)) - \sigma_t^2|\mathcal{F}_{t-1}] \\ &= Y_{t-1}. \end{aligned}$$

Then the sequence $\{Y_t : t = 1, 2, 3, \dots\}$ is a martingale. This can be used to show the distance a node has moved varies roughly between plus and minus the square root of the S_t .

For simplicity, we here let $\tau = T_{up}$. Since τ is a stopping time, by appropriately invoking the Optional Stopping Theorem, we obtain

$$E[Y_\tau|\mathcal{F}_{t-1}] = Y_0 = 0,$$

therefore,

$$E[(L_b(0) - L_b(\tau))^2] = S_\tau.$$

Also, according our definition, when the packet is delivered we have

$$0 \leq L_b(\tau) \leq r,$$

therefore,

$$(L_b(0))^2 \geq S_\tau \geq (L_b(0) - r)^2.$$

If $v_t, t = 1, 2, 3, \dots$ are identically and independently distributed, $\sigma_t^2 = \sigma^2$ for all t , then

$$\frac{(L_b(0))^2}{\sigma^2} \geq E[T_{up}] \geq \frac{(L_b(0) - r)^2}{\sigma^2}.$$

2) *Biased Packet Mobility*: The packet mobility is said to be *biased* if the corresponding sequence of random variable $L_b(t)$, $t \geq 1$, satisfies

$$E[L_b(t)|\mathcal{F}_{t-1}] < L_b(t-1).$$

Similarly, we have

$$E[v_t|\mathcal{F}_{t-1}] < 0.$$

Similarly, we are able to derive the theoretical bounds of the expected access time under biased mobility model.

Theorem 2: Given any packet mobility model that is biased, the expected access time is lower bounded by $\frac{L_b(0)-r}{-v}$, where $v < 0$.

Proof: Let $L'_b(t) = L_b(t) - H_t$, where $L_b(t)$ is a biased and $H_t = \sum_{i=1}^t E[v_i|\mathcal{F}_{i-1}]$.

$$\begin{aligned} E[L'_b(t)|\mathcal{F}_{t-1}] &= E[L_b(t) - H_t|\mathcal{F}_{t-1}] \\ &= E[L_b(t) - E[v_t|\mathcal{F}_{t-1}] - H_{t-1}|\mathcal{F}_{t-1}] \\ &= L_b(t-1) - H_{t-1} \\ &= L'_b(t-1). \end{aligned}$$

Then the sequence $\{L'_b(t) : t = 1, 2, 3, \dots\}$ is a martingale.

By using Optional Stopping Theorem, we have

$$E[L_b(0) - L_b(\tau)] = -H_\tau.$$

Also, according to our definition, when the packet is delivered we have

$$0 \leq L_b(\tau) \leq r,$$

therefore

$$L_b(0) \geq -H_\tau \geq L_b(0) - r.$$

If $v_t, t = 1, 2, 3, \dots$ are identically and independently distributed, $E[v_t|\mathcal{F}_{t-1}] = v, v < 0$ for all t , then

$$\frac{L_b(0)}{-v} \geq E[T_{up}] \geq \frac{L_b(0) - r}{-v}.$$

Remark 1: The above Theorems clearly indicate that finite network size have significant impact on the access time. Especially, in unbiased packet mobility case, the expected access time is proportional to the area of the square region, while the delay scales linearly with the initial distance from the source node to the nearest β -node if the packet mobility is biased. Our results show that biased packet mobility can achieve better delay performance than the unbiased packet mobility model. This is because the packet is inclined to move closer to the β -node. The model used in our analytical study is sufficiently general to encapsulate many popular mobility models (e.g., random way-point, random walk, Brownian motion).

IV. BIASED ROUTING SCHEME

Note that biased packet mobility has better delay performance than unbiased packet mobility. In this section, we will exploit the forwarding algorithms to achieve better delay performance while keep low resource usage compared to epidemic routing scheme. We assume at the beginning of every time slot, the nodes will estimate the distance to the nearest β -node and the step length, e.g. the distance it has moved towards the centering β -node (this can be done by measuring the signal strength from the centering access point β). Based on this local information, we are able to perform biased random walk in the MHNs.

In order to proceed, we need the following Lemma and Definition.

Lemma 1: Let us assume that a packet is currently with a node i at distance l_i from its destination, and that another node j is encountered. Then, the probability $l_i \rightarrow l_j$ that j is closer to the destination (i.e. packet gets forwarded) and is at distance l_j is equal to

$$P_{tx}(l_i \rightarrow l_j) = \sum_j \sum_{l_i > l_j} \frac{P(l_j)}{P(|X_i - X_j| \leq r|l_j)},$$

where $P(l_i)$ is the probability there exists at least one node which is l_i away from the β -node.

Proof: We know that node i is at distance l_i . j 's distance may have any value l_j with probability $P(l_j|l_i) = P(l_j|X_i - X_j| \leq r)P(|X_i - X_j| \leq r)/P(l_i)$ (Bayes rule). We also know that j 's distribution and we're interested in cases where $l_i > l_j$ (since only then is the packet forwarded). Finally, we know that j is within range of i (i.e., $|X_i - X_j| \leq r$). Therefore, the probability that j is at position X_j , given its distance to the destination l_j and the fact that it is within range of i is equal to

$$\frac{P(l_j)}{P(|X_i - X_j| \leq r|l_j)},$$

where

$$P(|X_i - X_j| \leq r|l_j) = \sum_{l_j: |X_j - X_i| \leq r} P(l_j).$$

Putting all these together, we obtain the probability of a transmission jumping from the state l_i to state l_j . Note that we have assumed that a packet get forwarded as long as the next hop has a smaller distance from the destination. ■

Suppose an "unfair" or "biased" random walk, with probability p it will move 1 step closer to β -node, and probability $q = 1 - p$ it will move 1 step away from the β -node.

Definition 2: (Packet's Biased Random Walk) A random walk has bias level p with respect to a given destination, and it is called p -biased random walk, if after a step the packet gets closer to the destination with probability p .

Let us start by considering a p -biased random walk started from $L_b(0)$ over the line $[0, \dots, 2l]$. For the simplicity, we scale and normalize the speed v to 1. Without loss of generality, let r be the destination. At each time slot, the packet moves closer to

the destination with probability p and away from the destination with probability $1-p$. The extreme points are reflective barriers.

To compute the expected access time $E[T_{BR}]$ of biased routing scheme, we use the method illustrated as below. The access time can be written as

$$E[T_{BR}] = \sum_{k=L_b(0)}^r h_p(k, k-1),$$

while $h_p(k, k-1)$ is the hitting time of node j starting from node i for a p -biased random walk. It is one less than the expected return time at position k of a p -biased random walk over the segment $[0...k]$. The return time is the inverse of the stationary probability of observing the packet at position k ; thus, if $\pi_{i,k}$ is the stationary probability that a packet over $[0, k]$ is observed at position i , we have

$$h_p(k, k-1) = \pi_{k,k}^{-1} - 1,$$

and

$$E[T_{BR}] = \sum_{k=r+1}^{L_b(0)} [\pi_{k,k}^{-1} - 1].$$

The probability $\pi_{k,k}$ can be obtained by solving an elementary Discrete Time Markov Chain (DTMC) with state space $[0, k]$ and transition probabilities

$$p_{i,i+1} = p, p_{i,i-1} = 1-p, 1 \leq i \leq k-1.$$

From the balance equation of such a DMTC, we can write

$$\pi_{k,k}^{-1} = \pi_{0,k}^{-1} \left(\frac{1-p}{p} \right)^{k-1},$$

and

$$\pi_{0,k}^{-1} = 1 + \frac{p^{k-1}}{(1-p)^{k-1}} + \frac{1 - \frac{p^{k-1}}{(1-p)^{k-1}}}{2p-1},$$

hence,

$$\pi_{k,k}^{-1} = 1 + \frac{(1-p)^{k-1}}{p^{k-1}} + \frac{\frac{p^{k-1}}{(1-p)^{k-1}} - 1}{2p-1}.$$

By substituting the above probability, we obtain

$$E[T_{BR}] = \sum_{k=r+1}^{L_b(0)} \left[1 + \frac{(1-p)^{k-1}}{p^{k-1}} + \frac{\frac{p^{k-1}}{(1-p)^{k-1}} - 1}{2p-1} \right],$$

that after some manipulation can be rewritten as

$$E[T_{BR}] = 1 + L_b(0) - r(2p-1) + 2 \left(\frac{p-1}{2p-1} \right)^2 \left[\left(\frac{1-p}{p} \right)^{L_b(0)-1} - 1 \right],$$

Under a formal point of view, the hitting time $E[T_{BR}]$ is lower than the approximate value $\frac{L_b(t)-r}{2p-1}$ for p such that $0.5 < p \leq 1$, as well as the unbiased value, $(L_b(0) - r)^2 / \sigma^2$. Now

$$(L_b(0) - r) / (2p-1) > (L_b(0) - r)^2,$$

until $(L_b(0) - r)$ reaches the value $l' = 1/(2p-1)$. This is a crossover point. For $L_b(0) - r < l'$,

$$(L_b(0) - r) / (2p-1) - E[T_{BR}] > (L_b(0) - r)^2 - E^p[T_{up}],$$

and then $E^p[T_{up}]$ is closer to $(L_b(0) - r)^2$, while for $L_b(0) - r > l'$, $E^p[T_{up}]$ is closer to $(L_b(0) - r) / (2p-1)$.

Now, using the function $(L_b(0) - r) / (2p-1)$, we can see that the speed becomes $(2p-1)$. This value, $2p-1$, is interpreted as the characteristic speed. Clearly, our biased routing scheme result in a biased packet mobility, and thus better delay performance.

Remark 2: From above, we have shown that there exists a phase transition phenomenon when we adopt the biased routing scheme. In this scheme, the current custodian will only forward the packet to the node which is closer to the β -node. This local forwarding decision could be based on the received signal strength from the β -nodes, which will send the periodic beacon signals to guide the mobile users. In this way, the expected access time can achieve linearity performance in large-scale network where the intermeeting time does not scale very well with the network size.

V. SIMULATION RESULTS

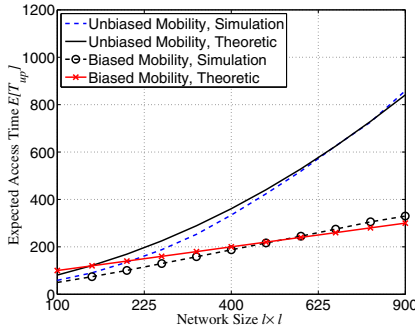
In this section, we compare our analytical results, regarding the expected access time of different packet mobility models, to simulation results.

We have used MatLab to evaluate the performance of different packet mobility models under a large range of connectivity levels, and for other mobility models (e.g., random walk). Although the networks of interest are disconnected in general, they may range from extremely sparse to almost connected networks, depending on the application. Connectivity is directly related to the number of forwarding opportunities presented to the packet custodian, and thus is expected to have an important effect on performance.

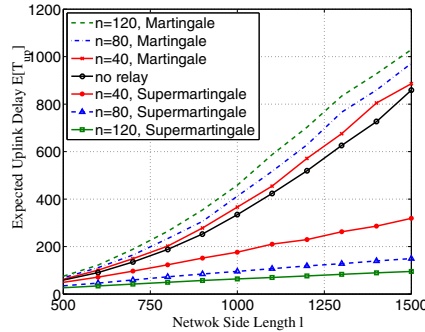
In all scenarios there are n α -nodes moving on a $l \times l$ two-dimensional torus (we choose torus for symmetry; we have run simulations that show that performances in bounded networks is similar), with a β -node placed at the center of the network. Each α -node is moving independently according to a certain mobility model. For simplicity, we scale the network size and make the average step length of a mobile node as 1. Each α -node can transmit up to distance $r \geq 0$, where r/l is smaller than the value required for connectivity, and each packet transmission takes one time slot. All results are averaged over 10000 runs.

In Figure 2(a), we fix $r = 1$, and compare analytical and simulation results for one-hop access time, for increasing the network size $l \times l$. Both analytical bounds and simulation results for the expected access time of unbiased and biased mobility models are included. As can be seen in Figure 2(a), our theoretic lower bounds hold pretty well. It has been shown clearly that the access time increases as network size increases, and the delay performance of unbiased mobility is much worse than that of biased mobility, especially in large-scale networks.

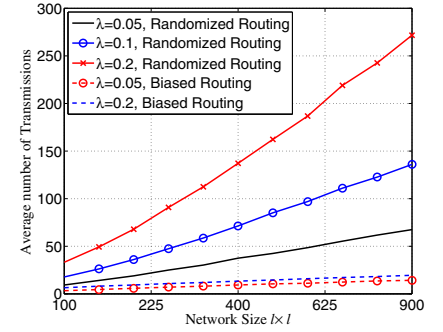
In Figure 2(b), we compared the expected access time with different routing schemes, and therefore different mobility models. We fix $r = 1$. When the node performs randomized routing scheme (unbiased mobility), i.e., forwards the packet to a randomly-chosen relay node within its communication range at the beginning of every time slot, the expected access time



(a) Theoretic results vs. Simulations.



(b) Expected access time.



(c) Number of Transmissions.

can be reduced by a constant. However, it still scales poorly with the network size. Nevertheless, if we choose the relay nodes from the forwarding set based on their current distance to the β -node, we show that there exists a phase transition such that the expected access time will be able to achieve linearity performance with respect to l (biased mobility). Also, as can be seen in this figure, when we increase the number of nodes in this network, the delay performance of both algorithms improves due to a higher chance to exchange packets with other nodes.

Finally, in Figure 2(c), we compare the average number of transmissions needed for a successful packet delivery under different schemes. The network size is fixed to 20×20 . As is evident by this figure, the average number of transmissions of randomized routing scheme is several times larger than that of proposed biased routing scheme. And the number of transmission increases as node density goes up. This is because the packet is more likely to be forwarded to another relay node when the network becomes denser. Therefore, we have shown that the biased routing scheme can achieve better delay performance while keeping the resource usage low. This is because this scheme can predict the gains and losses of the forwarding decisions more precisely with the position information of the possible relay candidates.

VI. CONCLUSION

In this paper, we have studied theoretical bounds of the access time in MHNs, which are helpful to increase our understanding of the fundamental properties and performance limits of MHNs, as well as to evaluate and optimize the performance of specific routing algorithms. Our contributions are threefold. First, we introduce the random process of packet mobility in mobile hybrid networks. Second, by using this new probabilistic model, we derive the lower bound of the access time for any forwarding algorithm, which depends on the network size and packet mobility models. Particularly, we have shown that, when wireless nodes move at an average speed of v and variance σ^2 towards an AP, the expected access time is lower bounded by $\frac{(L_b(0)-r)^2}{\sigma^2}$ if packet mobility is unbiased and $\frac{L_b(0)-r}{v}$ for the biased mobility, where $L_b(0)$ is the distance from the source node to the nearest access point. Third, we

proposed a biased routing scheme to minimize the access time such that data packets can be forwarded with lower resources usage and overhead. We also performed simulations to show the validity of our analytical results.

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